

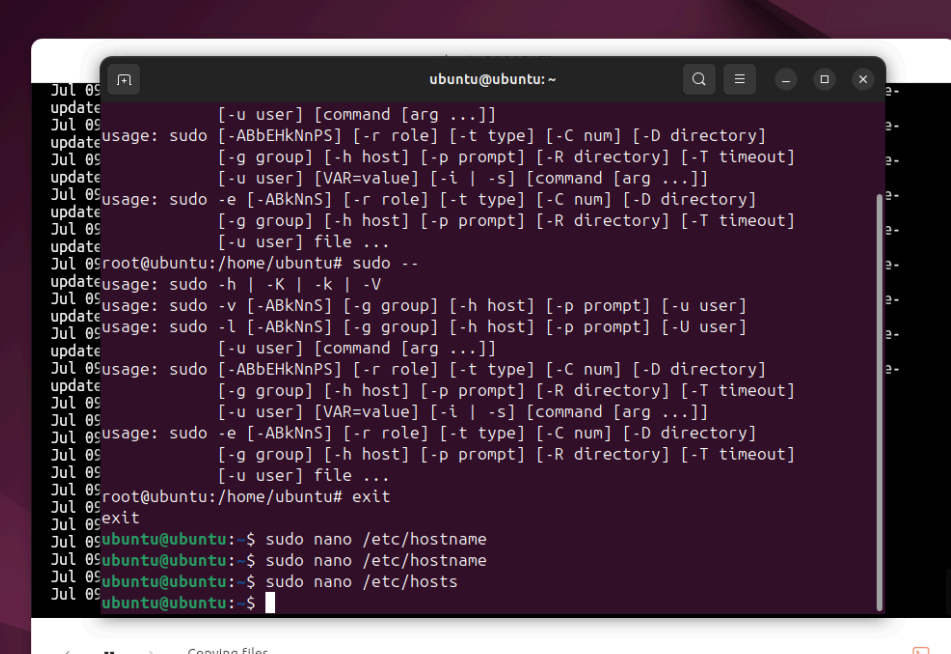


COURSE CODE/TITLE: CPE212 Automating Server Management	SECTION: CPE31S2
DATE PERFORMED: 07/09/2026	DATE SUBMITTED: 07/09/2026
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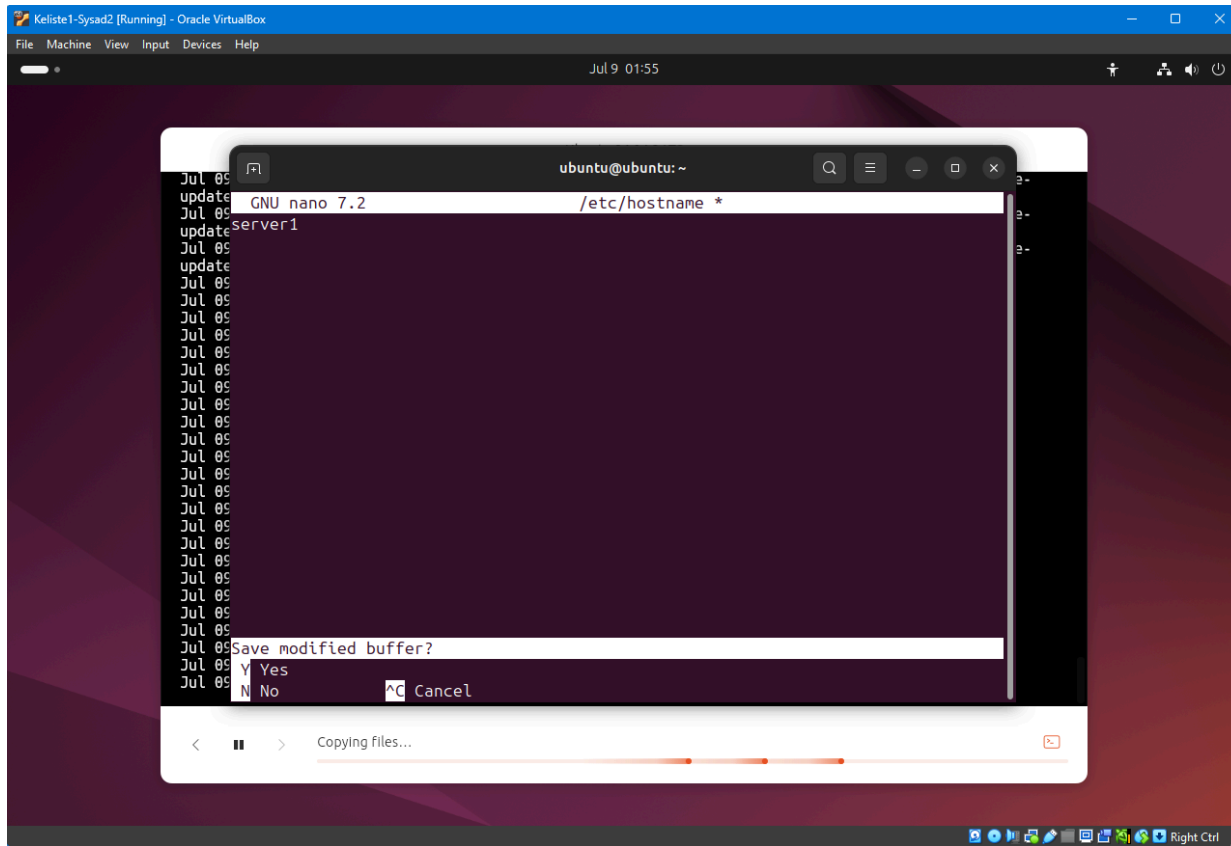
ACTIVITY NO. 1.1 Creating Virtual Machines

1. PROCEDURE OUTPUT

TASK 1:
Server 1



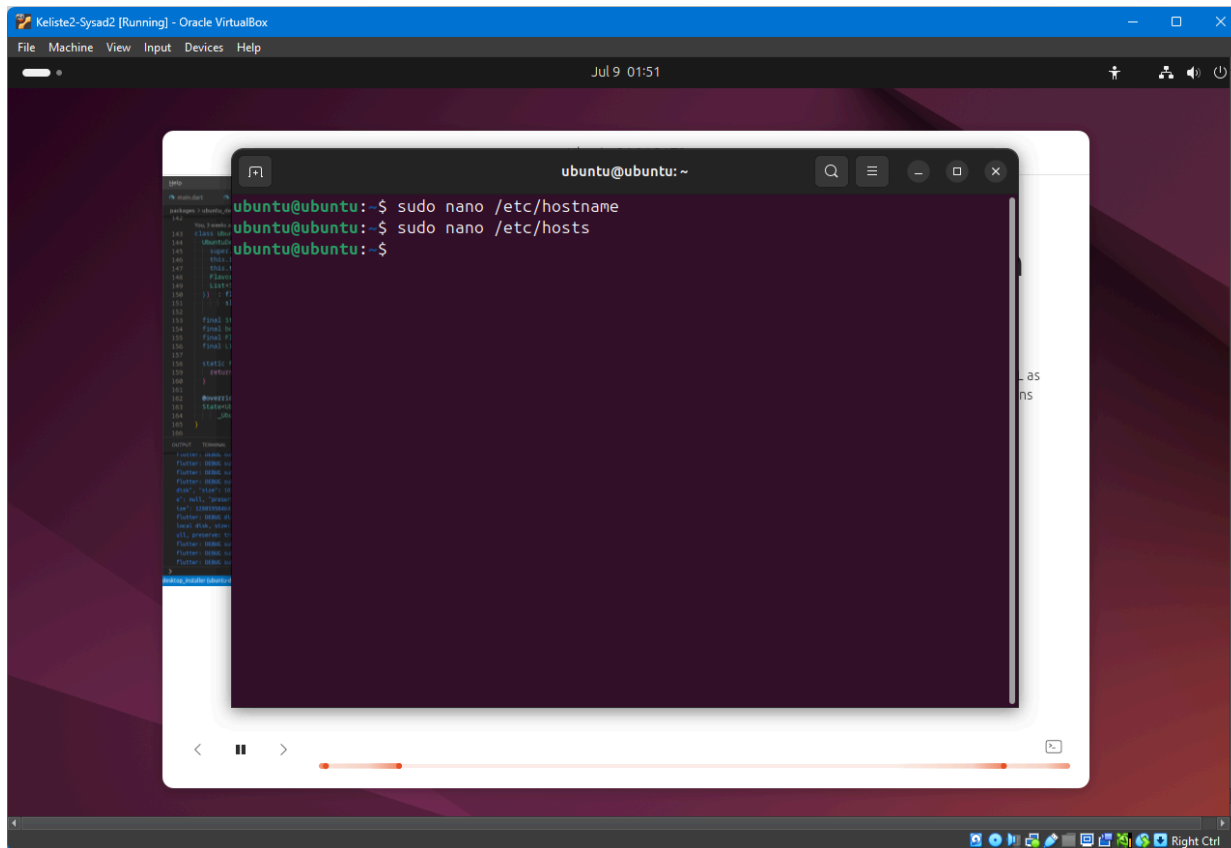
```
Jul 09 01:32
File Machine View Input Devices Help
Jul 09 01:32
ubuntu@ubuntu: ~
[ -u user ] [ command [ arg ... ] ]
Jul 09 update [ -ABbEHkNnPS ] [ -r role ] [ -t type ] [ -C num ] [ -D directory ]
Jul 09 update [ -g group ] [ -h host ] [ -p prompt ] [ -R directory ] [ -T timeout ]
Jul 09 update [ -u user ] [ VAR=value ] [ -i | -s ] [ command [ arg ... ] ]
Jul 09 update [ -e [ -ABkNnS ] [ -r role ] [ -t type ] [ -C num ] [ -D directory ]
Jul 09 update [ -g group ] [ -h host ] [ -p prompt ] [ -R directory ] [ -T timeout ]
Jul 09 update [ -u user ] file ...
Jul 09 root@ubuntu:/home/ubuntu# sudo --
Jul 09 update [ -h | -K | -k | -V ]
Jul 09 update [ -v [ -ABkNnS ] [ -g group ] [ -h host ] [ -p prompt ] [ -u user ]
Jul 09 update [ -l [ -ABkNnS ] [ -g group ] [ -h host ] [ -p prompt ] [ -U user ]
Jul 09 update [ -u user ] [ command [ arg ... ] ]
Jul 09 update [ -ABbEHkNnPS ] [ -r role ] [ -t type ] [ -C num ] [ -D directory ]
Jul 09 update [ -g group ] [ -h host ] [ -p prompt ] [ -R directory ] [ -T timeout ]
Jul 09 update [ -u user ] [ VAR=value ] [ -i | -s ] [ command [ arg ... ] ]
Jul 09 update [ -e [ -ABkNnS ] [ -r role ] [ -t type ] [ -C num ] [ -D directory ]
Jul 09 update [ -g group ] [ -h host ] [ -p prompt ] [ -R directory ] [ -T timeout ]
Jul 09 update [ -u user ] file ...
Jul 09 root@ubuntu:/home/ubuntu# exit
Jul 09 exit
Jul 09 ubuntu@ubuntu:~$ sudo nano /etc/hostname
Jul 09 ubuntu@ubuntu:~$ sudo nano /etc/hosts
Jul 09 ubuntu@ubuntu:~$ sudo nano /etc/hosts
Jul 09 ubuntu@ubuntu:~$
```

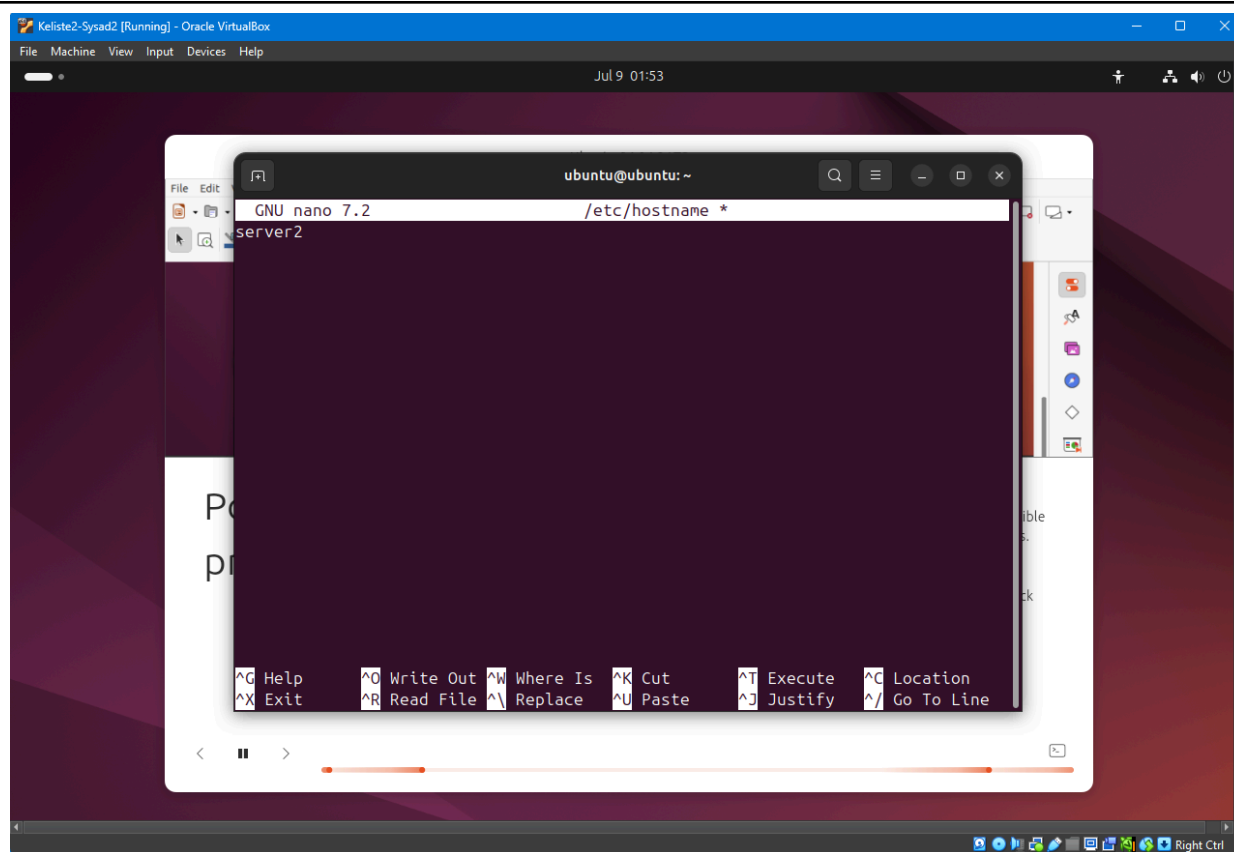


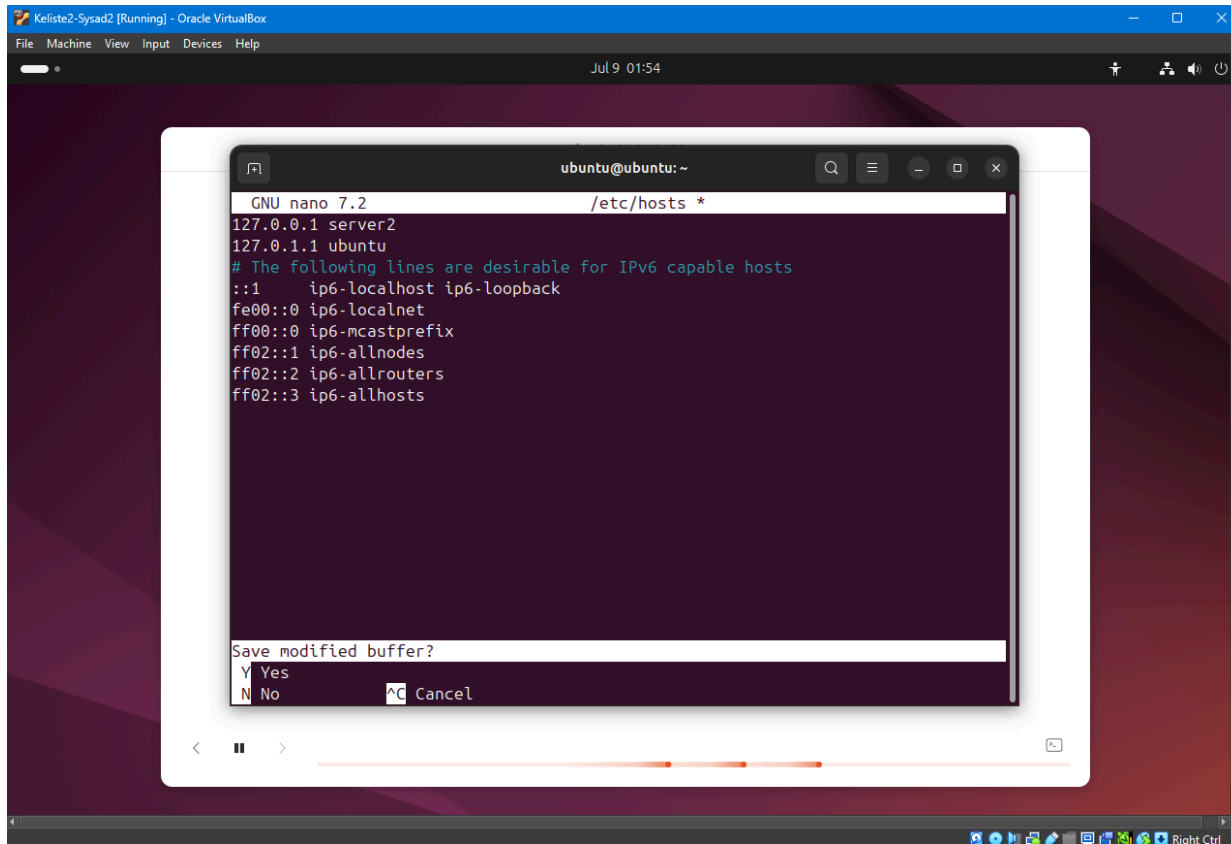
The screenshot displays a Kali Linux desktop environment. At the top, a terminal window titled 'Kali1-Sysad2 [Running] - Oracle VM VirtualBox' is visible. The terminal shows the user 'ubuntu@ubuntu: ~' editing the file '/etc/hosts' using 'GNU nano 7.2'. The file content includes IP addresses for 'server1' and 'ubuntu', followed by IPv6 addresses and their corresponding hostnames. The terminal output shows the user pressing 'Ctrl+O' to write the file and 'Ctrl+X' to exit, resulting in a confirmation message: 'File Name to Write: /etc/hosts'.

Below the terminal window, a file manager window is open, showing a progress bar for 'Copying files...'. The file manager interface includes a sidebar with 'Files' and 'Recent' tabs, and a main pane displaying a list of files and folders. The files listed include '127.0.0.1 server1', '127.0.0.1 ubuntu', and several IPv6 addresses. The file manager also shows a 'Copy' button and a 'Cancel' button.

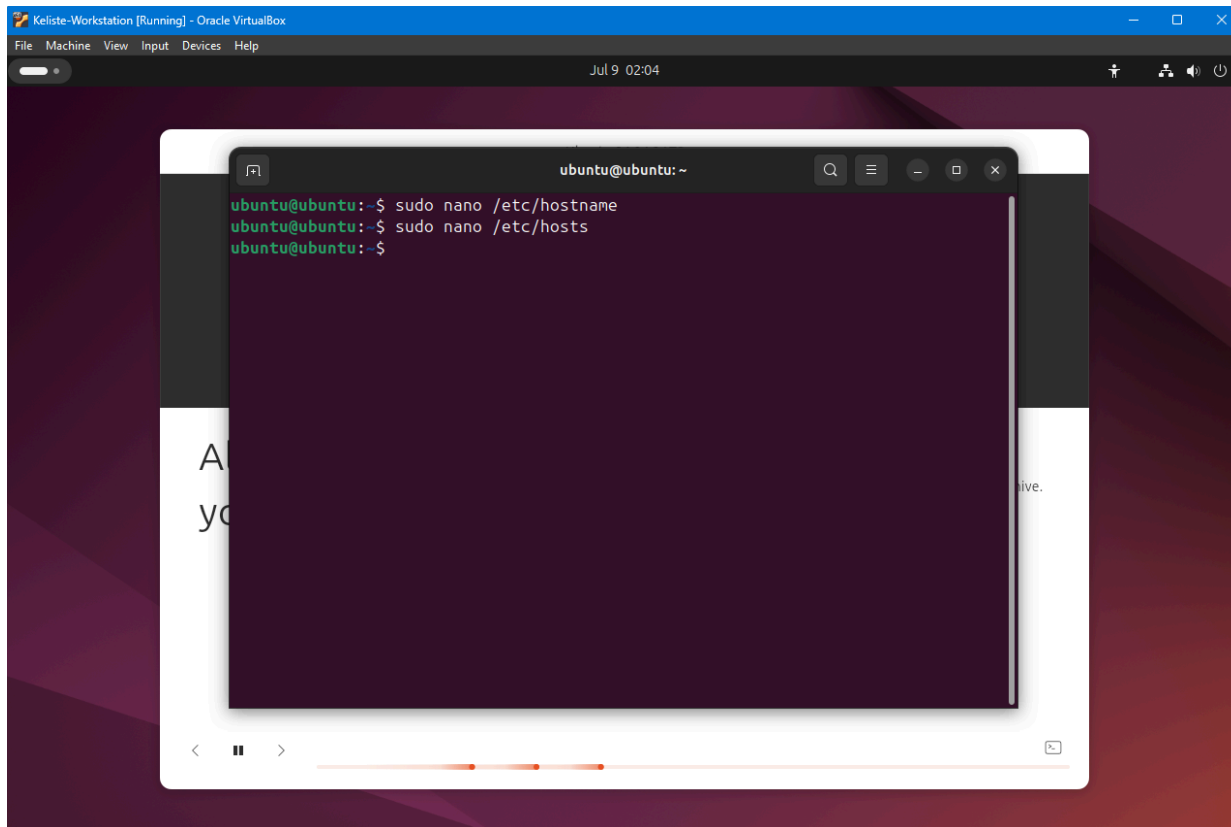
Server 2

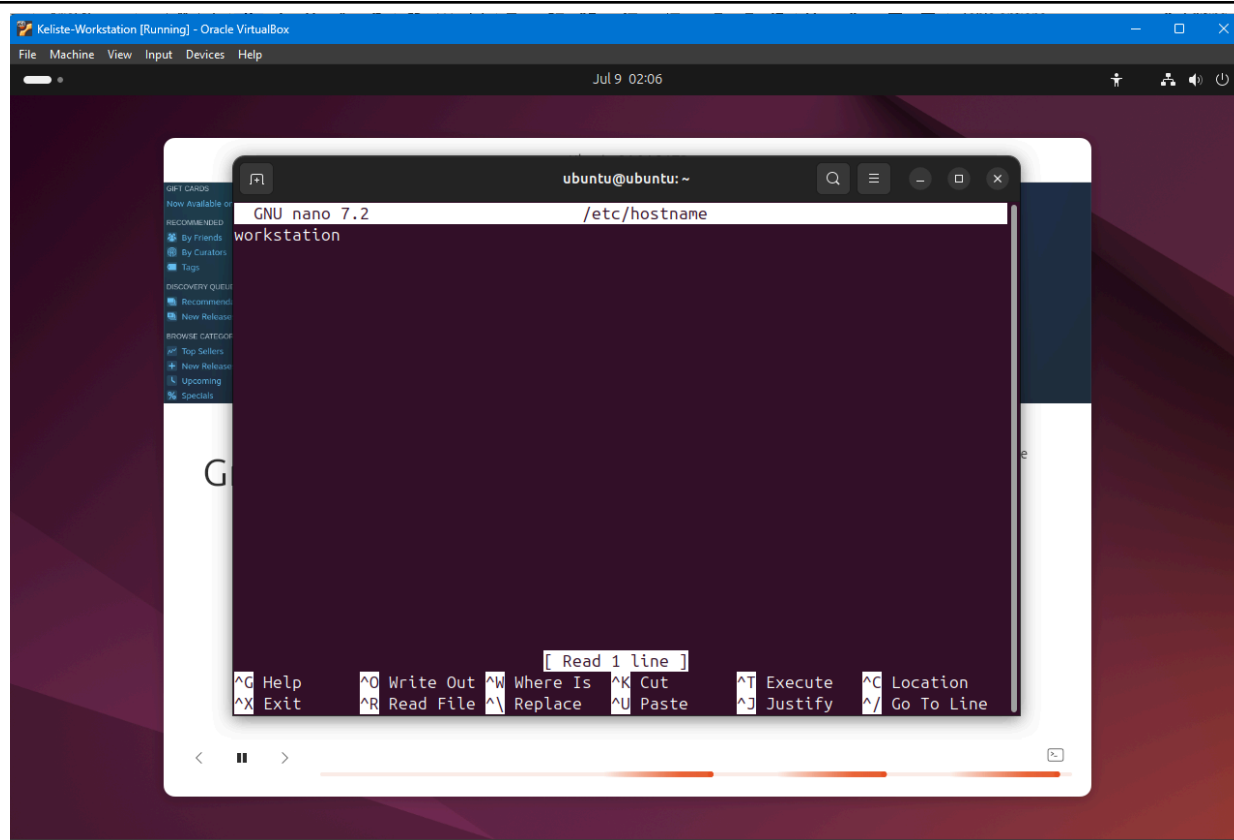


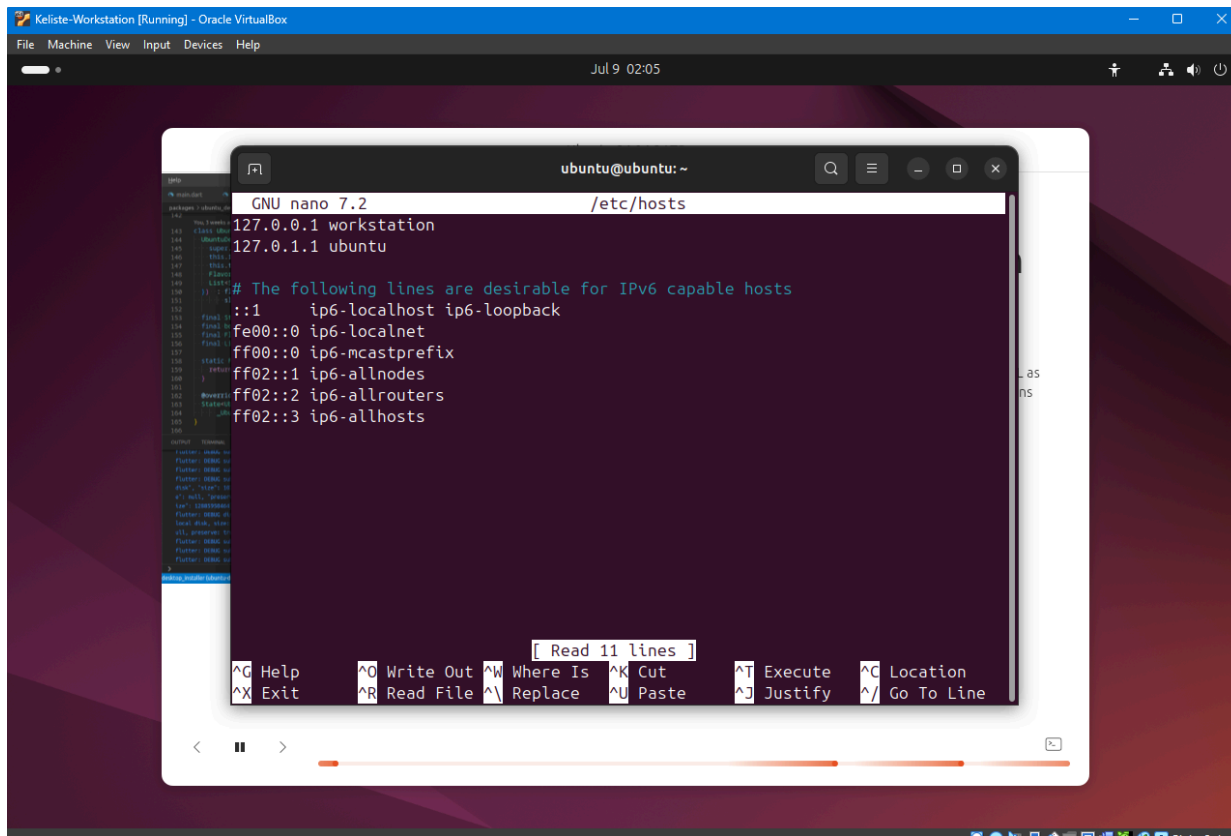




Workstation







TASK 2:

Server 1

The screenshot displays a Kali Linux desktop environment. The top panel shows the system menu with the text 'Kali Linux - Sysad2 [Running] - Oracle VirtualBox'. Below this is a menu bar with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. The main workspace contains two windows. The background window is a terminal titled 'ubuntu@ubuntu: ~' showing the output of the command 'sudo apt update'. The output lists various package updates from Ubuntu 24.04.2 LTS 'Noble Numbat', including 'noble-security', 'noble-updates', 'noble-backports', and 'noble-updates/main amd64 Packages'. A foreground window, likely a file manager, is partially visible at the bottom, showing a progress bar and the text 'Copying files...'.

```

Jul 09 08:00:00 ubuntu@ubuntu: ~
Jul 09 08:00:01 ubuntu@ubuntu:~$ sudo apt update
Jul 09 08:00:02 Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) nobl
Jul 09 08:00:03 e InRelease
Jul 09 08:00:04 Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) nobl
Jul 09 08:00:05 e Release
Jul 09 08:00:06 Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Jul 09 08:00:07 Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Jul 09 08:00:08 Get:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Jul 09 08:00:09 Get:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Jul 09 08:00:10 Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,079
Jul 09 08:00:11 kB]
Jul 09 08:00:12 Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [827
Jul 09 08:00:13 kB]
Jul 09 08:00:14 Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main i386 Packages [515 k
Jul 09 08:00:15 B]
Jul 09 08:00:16 Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [267
Jul 09 08:00:17 kB]
Jul 09 08:00:18 Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [18
Jul 09 08:00:19 kB]
Jul 09 08:00:20 Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main Icons (48x48) [37.3
Jul 09 08:00:21 kB]
Jul 09 08:00:22 Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main Icons (64x64) [52.7
Jul 09 08:00:23 kB]
Jul 09 08:00:24 Get:15 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata

```



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Keliste1-Sysad2 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Jul 9 02:09

```
ubuntu@ubuntu: ~  
Jul 09 02:05$ sudo apt update  
update:638 packages can be upgraded. Run 'apt list --upgradable' to see them.  
Jul 09 02:05$ sudo apt upgrade  
Jul 09 02:05$ Reading package lists... Done  
Jul 09 02:05$ Building dependency tree... Done  
Jul 09 02:05$ Reading state information... Done  
Jul 09 02:05$ Calculating upgrade... Done  
Jul 09 02:05$ The following packages were automatically installed and are no longer required:  
Jul 09 02:05$   libfwupd2 libgl1-amd-dri libglapi-mesa libllvm19 libwoff1  
Jul 09 02:05$ Use 'sudo apt autoremove' to remove them.  
Jul 09 02:05$ The following NEW packages will be installed:  
Jul 09 02:05$   libfwupd3 libllvm20 linux-headers-6.17.0-35-generic  
Jul 09 02:05$   linux-hwe-6.17.0-headers-6.17.0-35 linux-hwe-6.17.0-tools-6.17.0-35  
Jul 09 02:05$   linux-image-6.17.0-35-generic linux-modules-6.17.0-35-generic  
Jul 09 02:05$   linux-modules-extra-6.17.0-35-generic linux-tools-6.17.0-35-generic  
Jul 09 02:05$ The following upgrades have been deferred due to phasing:  
Jul 09 02:05$   gir1.2-mutter-14 libmutter-14-0 mutter-common mutter-common-bin  
Jul 09 02:05$ The following packages have been kept back:  
Jul 09 02:05$   libgl1-amd-dri  
Jul 09 02:05$ The following packages will be upgraded:  
Jul 09 02:05$   alsa-ucm-conf amd64-microcode apparmor apport apport-core-dump-handler  
Jul 09 02:05$   apport-gtk apt apt-utils avahi-daemon base-files bind9-dnsutils bind9-host  
Jul 09 02:05$   bind9-libs bluez bluez-cups bluez-obexd bsextrautils bsdutils  
Jul 09 02:05$   ca-certificates cifs-utils cloud-init coreutils cpp-13  
Jul 09 02:05$   cpp-13-x86-64-linux-gnu cryptsetup cryptsetup-bin cryptsetup-initramfs cups
```

< || > Copying files...

Right Ctrl

```

done.
Processing triggers for initramfs-tools (0.140ubuntu13.5) ...
update-initramfs: Generating /boot/initrd.img-6.8.0-134-generic
keliste@Keliste-Server1:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcb-1.0-0 libsbios-c2
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 752 kB of archives.
After this operation, 6,051 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-sftp-
server amd64 1:8.9p1-3ubuntu0.16 [38.9 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-serve
r amd64 1:8.9p1-3ubuntu0.16 [436 kB]

```

```

keliste@Keliste-Server1:~$ sudo service ssh start
keliste@Keliste-Server1:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: ena
   Active: active (running) since Thu 2026-07-23 09:03:54 +08; 50s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Main PID: 37062 (sshd)
    Tasks: 1 (limit: 4606)
   Memory: 1.7M
      CPU: 37ms
   CGroup: /system.slice/ssh.service
           └─37062 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 23 09:03:54 Keliste-Server1 systemd[1]: Starting OpenBSD Secure Shell serve>
Jul 23 09:03:54 Keliste-Server1 sshd[37062]: Server listening on 0.0.0.0 port 2>
Jul 23 09:03:54 Keliste-Server1 systemd[1]: Started OpenBSD Secure Shell server.
Jul 23 09:03:54 Keliste-Server1 sshd[37062]: Server listening on :: port 22.
lines 1-16/16 (END)

```

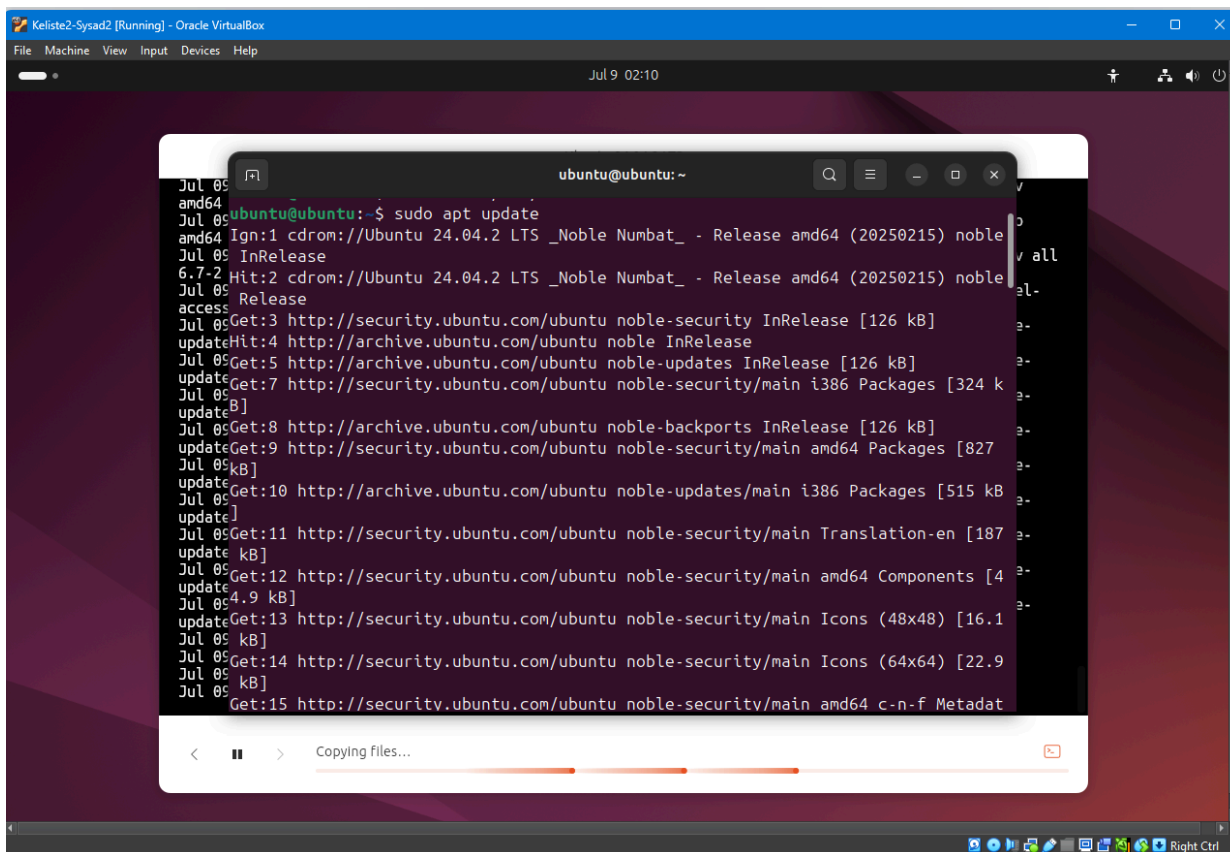
```
keliste@Keliste-Server1:~$ sudo ufw allow ssh

Rules updated
Rules updated (v6)
keliste@Keliste-Server1:~$ sudo ufw enable
Firewall is active and enabled on system startup
keliste@Keliste-Server1:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)

keliste@Keliste-Server1:~$
```

Server 2



The screenshot shows a VirtualBox window titled 'Keliste2-Sysad2 [Running] - Oracle VirtualBox'. Inside the window is an Ubuntu desktop environment. A terminal window is open, displaying the output of the command 'sudo apt update'. The output shows the system is updating from Ubuntu 24.04.2 LTS 'Noble Numbat'. It lists various updates including security updates, main packages, and backports, along with their sizes. A progress bar at the bottom of the terminal window indicates 'Copying files...'. The VirtualBox window's top menu bar includes 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. The status bar at the bottom right shows system icons and the text 'Right Ctrl'.

```
Jul 09 02:10:00 ubuntu@ubuntu:~$ sudo apt update
amd64 Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble
amd64 InRelease
Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:4 http://archive.ubuntu.com/ubuntu noble InRelease
Get:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [324 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [827 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main i386 Packages [515 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [187 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [4.9 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main Icons (48x48) [16.1 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/main Icons (64x64) [22.9 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata
```



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Keliste2-Sysad2 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Jul 9 02:11

```
ubuntu@ubuntu:~$ sudo apt upgrade
amd64 638 packages can be upgraded. Run 'apt list --upgradable' to see them.
Jul 09 02:11:01 ubuntu apt[1234]: Reading package lists... Done
amd64 6.7-2 Building dependency tree... Done
Jul 09 02:11:01 ubuntu apt[1234]: Reading state information... Done
amd64 Calculating upgrade... Done
Jul 09 02:11:01 ubuntu apt[1234]: The following packages were automatically installed and are no longer required:
update libfwupd2 libgl1-amber-dri libglapi-mesa libllvm19 libwoff1
Jul 09 02:11:01 ubuntu apt[1234]: Use 'sudo apt autoremove' to remove them.
Jul 09 02:11:01 ubuntu apt[1234]: The following NEW packages will be installed:
update libfwupd3 libllvm20 linux-headers-6.17.0-35-generic
Jul 09 02:11:01 ubuntu apt[1234]: linux-hwe-6.17-headers-6.17.0-35 linux-hwe-6.17-tools-6.17.0-35
update linux-image-6.17.0-35-generic linux-modules-6.17.0-35-generic
Jul 09 02:11:01 ubuntu apt[1234]: linux-modules-extra-6.17.0-35-generic linux-tools-6.17.0-35-generic
update The following upgrades have been deferred due to phasing:
update gir1.2-mutter-14 libinput-bin libinput10 libmutter-14-0 mutter-common
update mutter-common-bin
Jul 09 02:11:01 ubuntu apt[1234]: The following packages have been kept back:
update libgl1-amber-dri
Jul 09 02:11:01 ubuntu apt[1234]: The following packages will be upgraded:
update alsa-ucm-conf amd64-microcode apparmor apparmor-core-dump-handler
Jul 09 02:11:01 ubuntu apt[1234]: apparmor-gtk apt apt-utils avahi-daemon base-files bind9-dnsutils bind9-host
Jul 09 02:11:01 ubuntu apt[1234]: bind9-libs bluez bluez-cups bluez-obexd bsdxtrutils bsdxtrutils
Jul 09 02:11:01 ubuntu apt[1234]: ca-certificates cifs-utils cloud-init coreutils cpp-13
```

< || > Copying files...

Right Ctrl

```

keliste@Keliste-Server2: ~
keliste@Keliste-Server2:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcb-1.0-0 libsbios-c2
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 752 kB of archives.
After this operation, 6,051 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-sftp-
server amd64 1:8.9p1-3ubuntu0.16 [38.9 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-serve
r amd64 1:8.9p1-3ubuntu0.16 [436 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ncurses-term
all 6.3-2ubuntu0.2 [267 kB]
Get:4 http://ph.archive.ubuntu.com/ubuntu jammy/main amd64 ssh-import-id all 5.1
keliste@Keliste-Server2:~$ sudo service ssh start
skeliste@Keliste-Server2:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: ena
   Active: active (running) since Thu 2026-07-23 09:08:05 +08; 1min 2s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 37111 (sshd)
     Tasks: 1 (limit: 4606)
    Memory: 1.7M
       CPU: 28ms
    CGroup: /system.slice/ssh.service
            └─37111 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 23 09:08:05 Keliste-Server2 systemd[1]: Starting OpenBSD Secure Shell serve>
Jul 23 09:08:05 Keliste-Server2 sshd[37111]: Server listening on 0.0.0.0 port 2>
Jul 23 09:08:05 Keliste-Server2 sshd[37111]: Server listening on :: port 22.
Jul 23 09:08:05 Keliste-Server2 systemd[1]: Started OpenBSD Secure Shell server.
lines 1-16/16 (END)

```

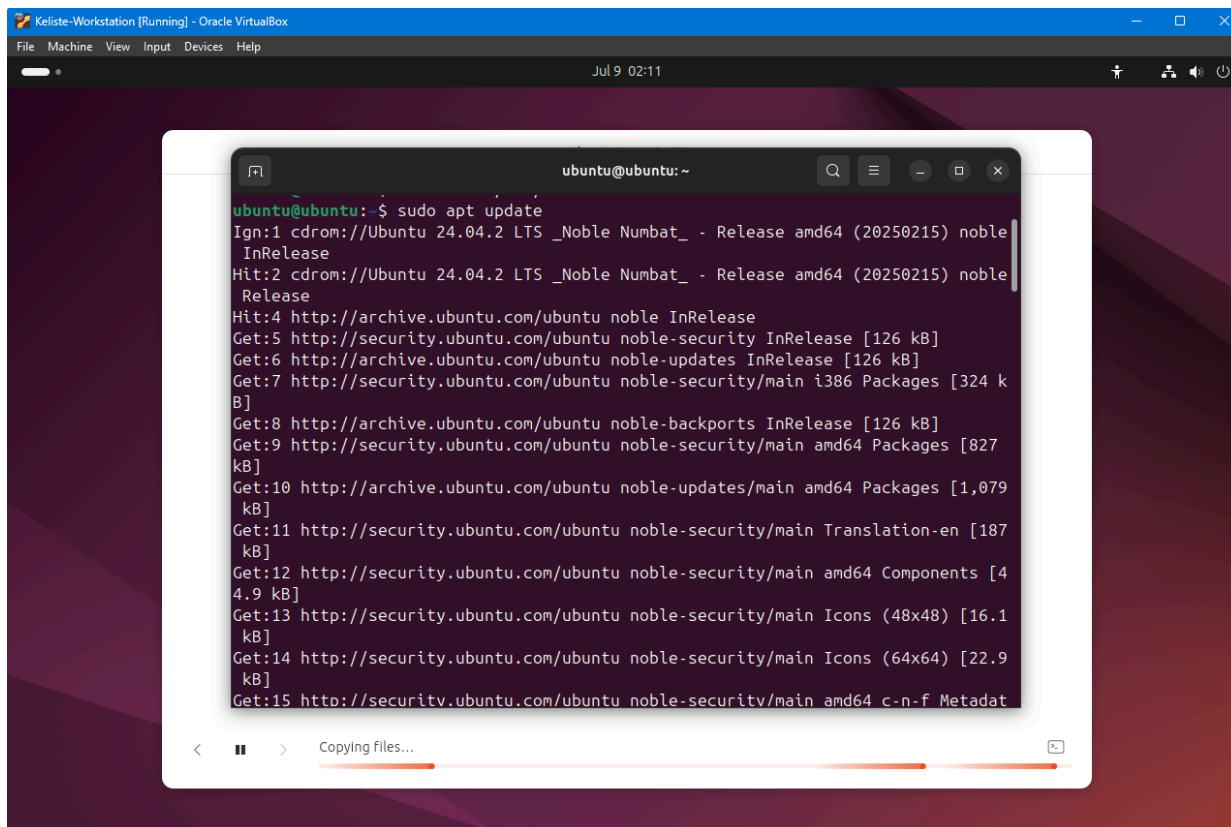


```
keliste@Keliste-Server2:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
keliste@Keliste-Server2:~$ sudo ufw enable
Firewall is active and enabled on system startup
keliste@Keliste-Server2:~$ sudo ufw status
Status: active
```

To	Action	From
--	-----	----
22/tcp	ALLOW	Anywhere
22/tcp (v6)	ALLOW	Anywhere (v6)

```
keliste@Keliste-Server2:~$
```

Workstation



Keliste-Workstation [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Jul 9 02:12

```

ubuntu@ubuntu:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done

The following packages were automatically installed and are no longer required:
  libfwupd2 libgl1-amber-dri libglapi-mesa libllvm19 libwoff1
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libfwupd3 libllvm20 linux-headers-6.17.0-35-generic
  linux-hwe-6.17-headers-6.17.0-35 linux-hwe-6.17-tools-6.17.0-35
  linux-image-6.17.0-35-generic linux-modules-6.17.0-35-generic
  linux-modules-extra-6.17.0-35-generic linux-tools-6.17.0-35-generic
The following upgrades have been deferred due to phasing:
  gir1.2-mutter-14 libmutter-14-0 mutter-common mutter-common-bin
The following packages have been kept back:
  libgl1-amber-dri
The following packages will be upgraded:
  alsa-ucm-conf amd64-microcode apparmor apport apport-core-dump-handler
  apport-gtk apt apt-utils avahi-daemon base-files bind9-dnsutils bind9-host
  bind9-libs bluez bluez-cups bluez-obexd bsdxtrutils bsdutils
  ca-certificates cifs-utils cloud-init coreutils cpp-13
  cpp-13-x86-64-linux-gnu cryptsetup cryptsetup-bin cryptsetup-initramfs cups
  
```

Copying files...

keliste@Keliste-Workstation: ~

```

update-initramfs: Generating /boot/initrd.img-6.8.0-134-generic
keliste@Keliste-Workstation:~$ sudo apt install openssh-server

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcb-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 752 kB of archives.
After this operation, 6,051 kB of additional disk space will be used.
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-sftp-server amd64 1:8.9p1-3ubuntu0.16 [38.9 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-server amd64 1:8.9p1-3ubuntu0.16 [436 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ncurses-term all 6.3-2ubuntu0.2 [267 kB]
  
```

```

keliste@Keliste-Workstation: ~
Setting up ssh-import-id (5.11-0ubuntu1) ...
Setting up ncurses-term (6.3-2ubuntu0.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...
keliste@Keliste-Workstation:~$ S
S: command not found
keliste@Keliste-Workstation:~$ sudo service ssh start
keliste@Keliste-Workstation:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enable
   Active: active (running) since Thu 2026-07-23 08:58:23 +08; 2min 0s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 39104 (sshd)
      Tasks: 1 (limit: 4606)
     Memory: 1.7M
        CPU: 43ms
    CGroup: /system.slice/ssh.service
            └─39104 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 23 08:58:23 Keliste-Workstation systemd[1]: Starting OpenBSD Secure Shell server>
Jul 23 08:58:23 Keliste-Workstation sshd[39104]: Server listening on 0.0.0.0 port 22.
Jul 23 08:58:23 Keliste-Workstation sshd[39104]: Server listening on :: port 22.
Jul 23 08:58:23 Keliste-Workstation systemd[1]: Started OpenBSD Secure Shell server.
lines 1-16/16 (END)

keliste@Keliste-Workstation:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
keliste@Keliste-Workstation:~$ sudo ufw enable
Firewall is active and enabled on system startup

keliste@Keliste-Workstation:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)

keliste@Keliste-Workstation:~$

```

Task 3:

- 1.1 Server 1 IP address: 192.168.56.103
- 1.2 Server 2 IP address: 192.168.56.104
- 1.3 Workstation IP address: 192.168.56.102

Server 1

```
keliste@Keliste-Server1:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::4da3:e0da:8d2d:9e86 prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:1a38:6a63:ef41:c48d prefixlen 64 scopeid 0x0<gl
obal>
    inet6 fd17:625c:f037:2:8599:bf50:4c00:f68b prefixlen 64 scopeid 0x0<gl
obal>
    ether 08:00:27:2a:90:87 txqueuelen 1000 (Ethernet)
    RX packets 96 bytes 20279 (20.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 170 bytes 21549 (21.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.103 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::a716:d98b:9ece:7ef7 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:79:ad:b5 txqueuelen 1000 (Ethernet)
    RX packets 92 bytes 16451 (16.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 44 bytes 6633 (6.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
keliste@Keliste-Server1:~$ ping -c 4 192.168.56.104
PING 192.168.56.104 (192.168.56.104) 56(84) bytes of data.
64 bytes from 192.168.56.104: icmp_seq=1 ttl=64 time=2.58 ms
64 bytes from 192.168.56.104: icmp_seq=2 ttl=64 time=0.851 ms
64 bytes from 192.168.56.104: icmp_seq=3 ttl=64 time=0.982 ms
64 bytes from 192.168.56.104: icmp_seq=4 ttl=64 time=0.913 ms

--- 192.168.56.104 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 0.851/1.332/2.584/0.724 ms
keliste@Keliste-Server1:~$
```

Server 2

```

keliste@Keliste-Server2: ~
keliste@Keliste-Server2:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:74d:3154:d04b:7dd3 prefixlen 64 scopeid 0x0<global>
    bal>
    inet6 fe80::bd54:6459:28b5:b216 prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:a6b8:b713:ba0:bbc5 prefixlen 64 scopeid 0x0<global>
    bal>
    ether 08:00:27:49:83:13 txqueuelen 1000 (Ethernet)
    RX packets 371 bytes 388070 (388.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 281 bytes 29798 (29.7 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.104 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::b6c5:99ff:1b28:7bc4 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:50:f9:76 txqueuelen 1000 (Ethernet)
    RX packets 76 bytes 15007 (15.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 71 bytes 9139 (9.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536

```

Workstation

```

keliste@Keliste-Workstation:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:d02f:797a:e7f:4056 prefixlen 64 scopeid 0x0<global>
    inet6 fd17:625c:f037:2:e417:bb03:80cd:f81f prefixlen 64 scopeid 0x0<global>
    inet6 fe80::e4f9:e14e:c990:1ebc prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:41:d0:f1 txqueuelen 1000 (Ethernet)
    RX packets 60 bytes 9233 (9.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 151 bytes 20223 (20.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.102 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::40b6:acb3:f886:bd75 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:4a:e7:c0 txqueuelen 1000 (Ethernet)
    RX packets 87 bytes 17096 (17.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 60 bytes 8142 (8.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)

```

```
keliste@Keliste-Workstation:~$ ping -c 4 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=2.08 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.957 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.968 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.652 ms

--- 192.168.56.103 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3009ms
rtt min/avg/max/mdev = 0.652/1.165/2.083/0.544 ms
keliste@Keliste-Workstation:~$ ping -c 4 192.168.56.104
PING 192.168.56.104 (192.168.56.104) 56(84) bytes of data.
64 bytes from 192.168.56.104: icmp_seq=1 ttl=64 time=2.06 ms
64 bytes from 192.168.56.104: icmp_seq=2 ttl=64 time=1.29 ms
64 bytes from 192.168.56.104: icmp_seq=3 ttl=64 time=0.933 ms
64 bytes from 192.168.56.104: icmp_seq=4 ttl=64 time=0.809 ms

--- 192.168.56.104 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3020ms
rtt min/avg/max/mdev = 0.809/1.271/2.056/0.485 ms
keliste@Keliste-Workstation:~$
```

Task 4:

```
keliste@Keliste-Workstation:~$ ssh keliste@192.168.56.103
The authenticity of host '192.168.56.103 (192.168.56.103)' can't be established.
ED25519 key fingerprint is SHA256:L8+xoAX1tr2m6AWwQnQDNP/E/x841EXcASYshAM2+rE.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.56.103' (ED25519) to the list of known hosts.
keliste@192.168.56.103's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

keliste@Keliste-Server1:~$

keliste@Keliste-Server1:~$ exit
logout
Connection to 192.168.56.103 closed.
```



```
keliste@Keliste-Workstation:~$ ssh keliste@192.168.56.104
The authenticity of host '192.168.56.104 (192.168.56.104)' can't be established.
ED25519 key fingerprint is SHA256:FiQxJc8PnOnmuV0eQBWd+/GDKdcVCeLNUjWkaRdlza8.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.104' (ED25519) to the list of known hosts.
keliste@192.168.56.104's password:
Permission denied, please try again.
keliste@192.168.56.104's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

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individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

keliste@Keliste-Server2:~$
```

```
keliste@Keliste-Server2:~$ exit
logout
Connection to 192.168.56.104 closed.
keliste@Keliste-Workstation:~$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

- How are we able to use the hostname instead of IP address in SSH commands?
 - In ssh it can resolve hostname because it will mapped the IP address in the lookup source, so the system translates the name to the IP before connecting.
- How secured is SSH?
 - SSH is secured because it uses a cryptographic algorithm for encryption. Protecting data like passwords and commands from being intercepted or read in transit.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Claude AI

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.



COMPUTER
ENGINEERING

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Purpose of Use: Grammar Correction, Idea Generation

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: Checking for some ideas and grammar correction

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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